

FINAL EXPLANATION: MATHEMATICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: MATHEMATICS GRADE 10 – SUB-STRAND 1.2 INDICES: FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

This Final Explanation is your chance to show how everything you learned in Lessons 1 to 8 fits together. Your task is to answer the driving question: How can writing numbers as powers and logarithms help us make sense of, and calculate with, the enormous and tiny quantities that appear in Kenyan farming, finance, engineering, and science? Before you write, look back at your Day-1 model, your notes, your Driving Question Board (DQB) questions, and your work on the Kitale maize doubling puzzle. Answer each section below in complete sentences, in the blank space under the prompt. Use specific evidence from the unit, such as the maize seed sequence 1, 2, 4, 8, ..., 256, the E. coli bacteria, the Richter scale, or your log table work. Do not just state a rule. Explain how and why it works, and show your working for any calculation. Use mathematical notation correctly, for example 2^8 , a^{m+n} and $\log_2 256$.

Section 1: Why write numbers as powers?

A smallholder farmer in Kitale plants 1 maize seed, and the total number of seeds doubles every season. The counts are 1, 2, 4, 8, 16, 32, 64, 128, 256, and by Season 20 the count exceeds 1,000,000. (a) Explain how you can write 256 using index notation, and what the base and the index each tell you. (b) Explain why index notation is more useful than writing out the number or the long string of multiplications when the farmer's numbers become very large. Refer to Season 20 in your answer.

Write your answer here:

Section 2: Why do the laws of indices work?

(a) Using the example $2^3 \times 2^4$, explain in your own words WHY we add the indices when multiplying powers of the same base. Expand the powers into repeated factors. (b) Using $2^5 \div 2^2$, explain why we subtract the indices when dividing. (c) Use the Quotient Law to explain why $a^0 = 1$ and why $2^{-3} = 1/8$. (d) Show how the Power Law lets you simplify $(2^3)^2$, and use the laws to work out $2^8 \times 2^4$ as a single power of 2. Explain how these laws connect to the maize doubling puzzle.

Write your answer here:

Section 3: From indices to logarithms

(a) In the maize puzzle you know there are 256 seeds. Explain how the question 'how many doublings produced 256 seeds?' leads to a logarithm, and write your answer using log notation. (b) Write $10^3 = 1000$ in logarithm form. (c) Explain why the laws of indices alone could not solve $2^x = 500,000$ when x is not a whole number, and describe what you can say about x using powers of 2 you know. (d) The 2016 Baringo earthquake had a Richter scale reading of 5.9. Explain what this number has to do with indices and why scientists find it useful.

Write your answer here:

Section 4: Calculating with logarithms

A trader needs to find 4680×327 (in KES) without a calculator. (a) Describe the steps you would follow using four-figure common logarithm tables, and name the logarithm law you use. (b) Carry out the calculation, showing the characteristic and mantissa of each logarithm, and give your answer to three significant figures. (c) Explain why this method is quicker and less error-prone than long multiplication, and how it links to the Product Law of indices.

Write your answer here:

Section 5: Answering the driving question

Now bring your thinking together. Write a connected explanation of the driving question: How can writing numbers as powers and logarithms help us make sense of, and calculate with, the enormous and tiny quantities that appear in Kenyan farming, finance, engineering, and science? Use at least **THREE** different pieces of evidence from the unit (for example the Kitale maize seeds, *E. coli* in a Nairobi laboratory, compound interest, SGR or geothermal quantities, or the Richter scale). Explain what indices do, what the laws allow you to do, where the index laws stop being enough, and how logarithms fill that gap. Finish by saying how your thinking has changed since your Day-1 model.

Write your answer here:

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Understanding of index notation and the laws of indices	Explains clearly WHY each law works using expanded factors (e.g. $2^3 \times 2^4 = 2^7$), covers the Product, Quotient, Power, Zero and Negative Index laws correctly, and uses notation accurately throughout.	States and applies most laws correctly with some reasoning from repeated factors; minor notation slips or one law only stated, not explained.	Lists rules with little or no explanation of why they work, or makes errors such as multiplying instead of adding indices.
Understanding of logarithms and their link to indices	Defines a logarithm as the index, converts accurately between index form and log form, and explains why logarithms are needed where the index laws stop (e.g. $2^x = 500,000$), including the Richter scale idea.	Converts between index and log form correctly and gives a basic reason why logarithms are needed; connection to the real-world scale is partial.	Treats logarithms as a separate unrelated topic, or makes conversion errors.
Accuracy of calculation with	Correctly identifies characteristics and	Follows the correct method with a small	Method is incomplete or incorrect, e.g.

logarithm tables	mantissas, adds logarithms, finds the antilog, gives an answer close to 1,530,000, and explains the link to the Product Law and why the method is efficient.	arithmetic or table-reading slip, or gives a weaker explanation of why the method works.	multiplies the logarithms instead of adding them, or no working shown.
Use of evidence and connection to the driving question	Uses at least three specific pieces of evidence from the unit (Kitale maize numbers, E. coli, Richter scale, etc.) with correct data, and builds a connected argument that answers the driving question and reflects on growth since the Day-1 model.	Uses two pieces of evidence accurately and addresses the driving question, with limited connection between ideas.	Little or no evidence from the unit, or evidence is inaccurate or not linked to the driving question.

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